

What a Backtest Overfitting Diagnostic Measures, and What Readers Take It to Mean

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Abstract

The probability of backtest overfitting measures how consistently the configuration selected in-sample continues to dominate its grid out-of-sample. Its authors state that the method assumes the grid represents the trials actually performed. Readers take a low value to mean that a backtest is not overfitted. These two readings coincide when the assumption holds and diverge when it does not, and we quantify the divergence.

Simulating a researcher who explores M configurations with no edge whatsoever and reports a grid of ten containing the winner, the selection damage grows from 0.066 to 0.141 as M goes from ten to a thousand, while the reported statistic falls from 0.54 to 0.10. The direction is the difficulty: a wider hidden search produces a winner that is an extreme of the whole sample and therefore dominates its grid more consistently, which is exactly what a low value records. The effect holds under the Sharpe ratio as originally specified and under the mean outcome, under selection on the full sample and on a training subperiod, with and without a genuine edge, and at every level of correlation between configurations we tested.

No statistic computed on the reported grid recovers what is missing. The diagnostic retains only the mean of the winner's out-of-sample ranks; their dispersion separates the two regimes when configurations are independent, and the separation vanishes once neighbouring configurations share sixty percent of their trades, as adjacent parameter values do.

A protocol condition does restore it. When the grid is fixed before evaluation, the statistic becomes insensitive to search outside it, with a trend of -0.00 against -0.38 under free search. But an unverified pre-declaration is worth little: a researcher who declares a grid and explores elsewhere anyway finds a better candidate outside it in eighty-eight percent of cases at a thousand external trials, and the damage returns. The condition that makes the diagnostic interpretable is therefore a timestamped record of the search, which is a documentary artefact rather than a property of the returns.

Keywords: backtest overfitting; cross-validation; multiple testing; pre-registration; research protocol; selection bias

JEL: C12; C52; G11; G17

1 Introduction

A quantitative researcher reports a backtest. To address the obvious objection, that the result was found by trying many things, the researcher computes the probability of backtest overfitting of [Bailey et al. \(2017\)](#) on the configuration grid and reports a low number.

The concern is long standing. [White \(2000\)](#) formalised data snooping across many models, and [Harvey et al. \(2016\)](#) argued that the resulting multiplicity requires a substantially higher bar for claiming a real predictor. The diagnostic of [Bailey et al. \(2017\)](#) sits in that lineage, alongside the deflated Sharpe ratio of [Bailey and López de Prado \(2014\)](#), which corrects a reported statistic for the number of trials undertaken.

The method is well specified. Given a matrix of returns for N configurations over T periods, combinatorially symmetric cross-validation splits the sample into S blocks, considers every partition into equal in-sample and out-of-sample halves, and records how often the configuration best in-sample ranks below the out-of-sample median. What it measures is therefore the consistency with which the in-sample winner continues to dominate its grid. Its authors are explicit that the input is the set of trials performed, and that omitting trials biases the result.

The assumption is rarely satisfied. Research does not proceed by declaring a grid and evaluating it once; it proceeds by exploration. By the time a grid is written into a paper, the winner is already known, and the grid is built to contain it.

What has not been measured is the size and direction of the resulting error. One might expect unreported search to make the statistic uninformative, drifting toward the value it takes when selection is pure noise. It does something else. It moves systematically toward the value that reassures, faster as the hidden search grows.

We make three contributions. We quantify the divergence between what the diagnostic measures and what a reader takes it to mean. We show that no summary of the reported grid recovers the missing information, reporting a natural candidate that fails. And we identify the protocol condition that restores interpretability, along with the reason a declaration alone does not satisfy it.

2 Design

We simulate a researcher, and we know the truth by construction.

Return generation. R-multiples are drawn on the support $\{+2, 0, -1\}$ with probabilities calibrated to a real foreign exchange trade sequence: mean 0.192, standard deviation 1.271. A target mean is imposed by shifting mass between the extreme points. Sample length is $T = 800$ throughout.

Correlation between configurations. Adjacent parameter values on a real grid produce overlapping trades. We control this with a sharing parameter ρ : each configuration takes a given trade from a common draw with probability ρ and from its own draw otherwise. Independent configurations correspond to $\rho = 0$, and we treat $\rho = 0.6$ as the realistic case.

The researcher. M configurations are generated, all with the same true edge μ . The one with the best in-sample performance is retained. When $\mu = 0$ every configuration is worthless, so any apparent performance in the winner is selection alone, and

$$\text{illusion} = \text{apparent performance of winner} - \text{true edge of winner}$$

is the entire selection damage, known exactly.

The reported grid. The researcher reports ten configurations: the winner and nine drawn at random from those explored. The diagnostic is computed on those ten, with $S = 8$ blocks and all seventy partitions.

Robustness. Every cell is run under two performance measures, the Sharpe ratio as [Bailey et al. \(2017\)](#) specify and the mean outcome; under two selection regimes, the full sample and a contiguous first half; and with and without a genuine heterogeneous edge. Figures are averages over one hundred and fifty replications.

3 The divergence

Table 1: Reported diagnostic and true selection damage as unreported search grows. No configuration has any edge. Selection on the full sample; reported grid of ten; independent configurations.

Configurations explored	Sharpe ratio		Mean outcome	
	Reported PBO	Illusion	Reported PBO	Illusion
10	0.54	0.063	0.51	0.066
50	0.32	0.094	0.33	0.093
200	0.19	0.117	0.19	0.119
1000	0.10	0.141	0.12	0.141

Table 1 shows the two columns moving in opposite directions. As the hidden search grows by two orders of magnitude, the damage it causes more than doubles while the statistic falls by a factor of five.

The value at $M = 10$ deserves comment. A reading near 0.5 is what the diagnostic returns when selection carries no information: the in-sample winner is as likely to fall below the out-of-sample median as above it. That is the correct signature of worthless configurations. The more the researcher searched, the further the statistic moves from that value, in the direction a reader takes as robustness.

Table 2: Same design with a genuine heterogeneous edge, true means drawn around 0.10 with dispersion 0.05. Selection is partly legitimate.

Configurations explored	Sharpe ratio		Mean outcome	
	Reported PBO	Illusion	Reported PBO	Illusion
10	0.25	0.044	0.25	0.048
200	0.05	0.082	0.05	0.082
1000	0.02	0.101	0.02	0.099

Table 2 is the case closest to practice, where some configurations genuinely are better. The divergence is sharper. At a thousand hidden trials the statistic reads 0.02, which reads as strong evidence of a robust backtest, while the reported performance overstates the winner’s true edge by 0.10, comparable to the edge itself.

3.1 Mechanism, and why this is not a defect

The diagnostic asks how consistently the in-sample winner dominates its grid out-of-sample. A winner drawn from ten candidates is barely distinguishable from the other nine; across the seventy partitions its rank is close to random, and the statistic lands near one half.

A winner drawn from a thousand is an extreme order statistic of the whole sample. Placed among nine ordinary configurations, it leads in almost every partition, in-sample and out-of-sample alike, because its realised path is exceptional throughout rather than only in the half that selected it. Consistent dominance is what a low reading records.

The statistic is therefore doing exactly what it was built to do. The winner really does dominate the grid it was reported with. What diverges is the inference a reader draws, because dominance over a reported grid and absence of overfitting are the same thing only when the grid is the search.

3.2 Robustness

Replacing the Sharpe ratio by the mean outcome changes the trend from -0.43 to -0.39 . Selecting on a contiguous first half rather than the full sample gives -0.20 rather than -0.43 , so the effect is present under both and stronger under the regime the method assumes. Under correlation the reported statistic stays between 0.10 and 0.12 across $\rho \in \{0, 0.3, 0.6, 0.8\}$ at a thousand hidden trials. None of these choices is responsible.

4 No statistic on the grid recovers it

Since the diagnostic retains only the mean of the winner’s out-of-sample ranks, a natural remedy is to use their dispersion, which the mean discards. The intuition is that genuine superiority produces stable dominance while a large-sample fluke produces uneven dominance.

With independent configurations the separation is clean. At a comparable reported value, a winner from a thousand hidden trials shows rank dispersion of 0.196 against 0.127 for a genuinely superior configuration, and the true illusion differs by a factor of five that the reported value misses entirely.

It does not survive correlation. Table 3 reports the gap as neighbouring configurations come to share trades.

Table 3: Rank dispersion under correlated configurations. ρ is the fraction of trades shared between neighbours.

ρ	Hidden search, no edge		Real advantage	
	PBO	Dispersion	Dispersion	Gap
0.0	0.10	0.184	0.115	+0.069
0.3	0.11	0.200	0.160	+0.040
0.6	0.12	0.202	0.203	−0.001
0.8	0.11	0.201	0.218	−0.017

The gap closes at $\rho = 0.6$ and reverses beyond. Since adjacent parameter values on a real grid share well more than sixty percent of their trades, the remedy does not apply where it would be needed. We report it because it closes a natural line of inquiry: the information the diagnostic lacks is absent from the reported grid, and no summary of that grid appears to recover it.

5 The protocol condition

If the missing information is not in the data, it must come from the process. We test the obvious candidate: fixing the grid before evaluation.

Table 4: Free search against pre-declared grid. No edge, $\rho = 0.6$, reported grid of ten. Under pre-declaration the grid is the search, so the column heading describes configurations that were never evaluated.

Configurations outside grid	Free search		Pre-declared	
	PBO	Illusion	PBO	Illusion
10	0.50	0.062	0.51	0.048
50	0.32	0.084	0.50	0.049
200	0.20	0.101	0.52	0.052
1000	0.12	0.112	0.51	0.051

Pre-declaration restores the diagnostic exactly. The trend is -0.00 against -0.38 under free search, and the reported value stays at the noise level of one half where it belongs when configurations are worthless. This is unsurprising once stated: under pre-declaration nothing happens outside the grid, so the statistic cannot depend on it.

5.1 An unverified declaration is not the condition

The interesting case is imperfect compliance, since a rule that works only under perfect compliance is of no practical use. We model a researcher who declares a grid, explores outside it anyway, and substitutes the external winner into the declared grid if it performs better.

Table 5: Imperfect compliance. The grid is declared but external search continues and the better candidate is substituted. No edge, $\rho = 0.6$.

Configurations outside grid	PBO	Illusion	Substitution rate
10	0.36	0.076	52%
50	0.32	0.087	73%
200	0.26	0.101	78%
1000	0.18	0.115	88%

The damage returns almost in full. At a thousand external trials the external winner is better in eighty-eight percent of cases, the reported statistic falls to 0.18, and the illusion is 0.115 against 0.051 under genuine compliance. Even at ten external trials, substitution occurs in half of cases, because a ten-configuration external search competes evenly with a ten-configuration grid.

The condition is therefore not that a grid be declared but that the declaration be verifiable. A timestamped record distinguishes the two regimes; an assertion in a paper does not.

6 A worked example

Two researchers work on the same universe of strategies, in which no configuration has any edge at all. Neither knows this. Both compute the diagnostic correctly.

Researcher A fixes a grid of twelve configurations, timestamps the file, then evaluates. She reports what she declared.

Researcher B explores four hundred configurations, retains the best, then composes a grid of twelve around it for publication.

Table 6: Two researchers, one draw. No configuration in either universe has any edge. $T = 800$, $\rho = 0.6$.

	Researcher A (grid declared)	Researcher B (search unreported)
Configurations evaluated	12	400
Configurations reported	12	12
Sharpe ratio	0.052	0.069
t -statistic	1.47	1.94
PBO	0.61	0.20
True edge	0.000	0.000

A reader comparing the two publications prefers B on every reported criterion. Over three hundred repetitions of the same scenario, B obtains the lower diagnostic in ninety-one percent of

cases, with a mean of 0.16 against 0.49, and reports a Sharpe ratio higher by 0.039 on strategies that are worthless in both cases.

Nothing in the published numbers separates them. Both calculations are correct, both grids contain twelve configurations, both diagnostics follow the method. The only difference is a timestamped file that one researcher holds and the other does not, and that file, rather than any property of the returns, is what makes the diagnostic interpretable.

7 What this does and does not establish

It does not establish that the diagnostic is wrong. It answers a stated question about a stated input, and its authors flag the assumption we violate. The winner genuinely does dominate the grid it was reported with.

It does not establish that any published result is overfitted. We simulate; we audit no one's work.

What it establishes is that the failure mode is not the benign one. Under unreported search the statistic does not become noisy or drift toward its uninformative value; it moves monotonically toward the value that reassures. It establishes that no summary of the reported grid recovers what is missing, since the natural candidate fails at realistic correlation. And it establishes that pre-declaration restores interpretability only when verifiable, since an unverified declaration leaves eighty-eight percent of the damage intact.

Two practical consequences follow. A reported value well below one half deserves more scrutiny than a value at it, not less, since a very low reading is what a large hidden search produces on worthless configurations. And a diagnostic of this family is interpretable only alongside a record of the search, which is a documentary requirement rather than a statistical one. The same objection applies to the deflated Sharpe ratio (Bailey and López de Prado, 2014), which takes the number of trials as a declared input that no property of the data can verify.

8 Limitations

The return process is a three-point distribution calibrated to one real sequence, with correlation induced by shared draws. Real configuration returns are correlated through a richer structure, and while we vary ρ across its full range, the mechanism generating that correlation is stylised.

The reported grid is drawn at random from the explored set. A researcher who assembles a grid deliberately, choosing neighbouring parameter values around the winner, may produce different behaviour; the realistic case is plausibly worse, since neighbours of a lucky configuration inherit part of its luck.

Our model of imperfect compliance is one specific behaviour, substituting an external winner into a declared grid. Other departures exist, notably adjusting the declared grid itself after the fact, and are not covered.

We test one grid size against four search sizes; the interaction between the two is not mapped.

Finally, the qualitative point that omitting trials biases the diagnostic is known and stated by its authors. The contributions here are the magnitude, the direction, the failure of the natural remedy, and the distinction between a declaration and a verifiable one.

9 Conclusion

A diagnostic computed on a grid assembled after the winner is known does not fail gracefully. Across four search sizes, two performance measures, two selection regimes, four correlation levels, and with or without a true edge, it falls as the damage it is read to exclude rises.

The reading is inverted. A researcher who searched widely and reported narrowly obtains a lower number than one who searched narrowly and reported honestly, and no statistic on the reported grid distinguishes them. Fixing the grid in advance does distinguish them, and only when the fixing can be checked.

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Code availability

Simulation code reproducing every table is available on request and will be posted alongside the working paper.

Disclosure of interest

The author is the founder of OrgaX LLC, a quantitative research firm. This paper uses simulation only and audits no third party's work.

Use of generative AI

A generative AI assistant was used to support code development and manuscript editing. The author reviewed all content and takes full responsibility for the manuscript.

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